Availability and Usage of Pre-trained ANNs: Benchmarking Interactive Multi-Criterion Decision-Making Procedures Following Machine Learning-Based Decision-Maker

Deepanshu Yadav, Kalyanmoy Deb
Michigan State University, MI, USA

1 Introduction

This document provides pre-trained ANN models– (i) class predictor ANN model ($\mathcal{F}_{CL}$) and (ii) parameter predictor ANN model ($\mathcal{F}_{PP}$), for the benchmark and real-world engineering problems discusses in the paper: "*Benchmarking Interactive Multi-Criterion Decision-Making Procedures Following Machine Learning-Based Decision-Maker*", published in **The Genetic and Evolutionary Computation Conference July 13 - 17, 2026**.

2 List of Problems Used in Paper

Table 1: Problem description and parameter setting for GA and Machine-DM. Parameters M, n, and J are number of objectives, variables, and constraints, respectively.

Examples	Problem Details			Machine-DM Parameters		ANN1/ANN2 Parameters		
	M	n	J	$\hat{\delta}$	$\hat{\alpha}$	lr_1	lr_2	l_{reg}
Knee (2-objective) [1]	2	5	0	0.100	0.075	0.0100	0.010	0.010
Knee (3-objective) [1]	3	5	0	1 0.050	0.100	0.0010	0.001	0.010
Crashworthiness [2]	3	5	0	0.100	0.050	0.0010	0.002	0.020
Car Side Impact [3]	3	7	10	0.100	0.075	0.0020	0.001	0.012
River Pollution [4]	4	2	0	0.015	0.025	0.0025	0.001	0.010
WATER Example [5]	5	3	7	0.025	0.035	0.0010	0.002	0.020
NIMBUS-study [6]	6	2	0	0.030	0.050	0.0020	0.001	0.010
8-objective DTLZ2 [7]	8	10	0	0.050	0.200	0.0010	0.001	0.010

3 A Big Picture of Machine-DM

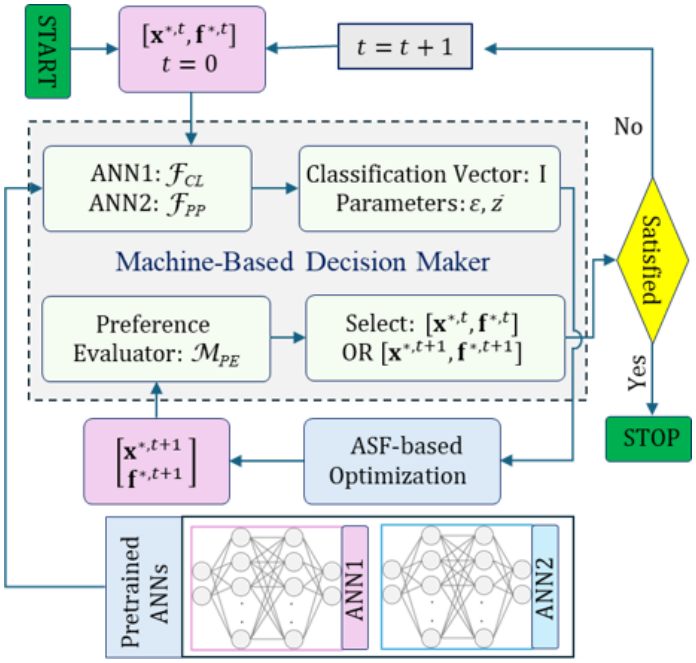

Figure 1: Illustration of Machine-DM approach integrated with an arbitrary iMCDM procedure.

The big picyure if Machine-DM integrated with an arbitrary iMCDM procedure is provided in Figure 1. In this document, we provide the pre-trained ANNs for each problems provided in Table 1. Information on the availability of the pre-trained ANNs are provided using this GitHub Link.

4 Instruction on Using ANN1 and ANN2

Please follow the step-by-step procedure:

1. Unzip the file– '`Pretrained ANNs.zip`'. It contains all the pre-trained ANNs for each problem provided in Table 1.

2. Open the unzipped 'Pretrained ANNs' folder and find subfolder for each problem.

3. Open the problem-specific subfolder, for example– '`3.Crash_3Obj`'.

4. Get the ANNs and scaler files, for example–

   - `Ex3_ANN1.keras`: Objective Class ($\mathbf{I}$) Predictor– ANN1 ($\mathcal{F}_{CL}$).
   - `Ex3_ANN2.keras`: Bounding Parameter ($\bar{z}, \varepsilon$) Predictor– ANN2 ($\mathcal{F}_{PP}$).
   - `Ex3_scaler_X.pkl`: To appropriately normalize the PO solution $\mathbf{x}^*$ for its use in ANN2 ($\mathcal{F}_{PP}$).

- `Ex3_scaler_Y.pkl`: To appropriately de-normalize the predicted bounding parameters $(\bar{z}, \varepsilon)$ for its use in ANN2 ($\mathcal{F}_{PP}$).

5. To implement Machine-DM for a given problem (say three-objective vehicle crashworthiness [2]), follow the python code provided in file '`Testing MachineDM as New User.pdf`' provided with this Zenodo link.

References

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